

MCQ & WRITTEN PHASE 6 REPORT

1. Executive Summary & Demographics

Following a clear data cleaning process, a total of 188 valid responses were analyzed. The participant pool represents a diverse academic background with a focus on technical fields.

- **Gender Distribution:** 119 Male participants; 69 Female participants.
- **Primary Field of Study:** Computer Science (CS) and related IT disciplines.
- **Academic Standing:** The collective average CGPA of the respondents is 3.01.

2. Examination System Evaluation

The research evaluated student perceptions regarding the two primary assessment formats: Multiple Choice Questions (MCQ) and Written Examinations.

- **Preference Metrics:** 91 respondents prefer MCQs, 66 prefer Written exams, and 31 expressed a preference for both formats.
- **Qualitative Strengths:** Respondents concluded that MCQs provide high fairness and objectivity. Conversely, Written questions were rated highly for depth of understanding and measuring comprehensive knowledge.

3. Stress Analysis and Distribution

A significant portion of the study focused on identifying primary causes of academic stress.

- **Primary Stressors:** 107 students cited Written exams as their primary cause of stress, compared to 55 for both formats and 26 for MCQs only.
 - **Demographic Stress Levels:** Analysis indicates that Post-Graduate students experience the highest stress levels, followed by Undergraduate and High School students respectively.
 - **Sentiment on Written Stress:**
 - Strongly Agree: 66
 - Agree: 48
 - Neutral: 43
 - Disagree: 19
 - Strongly Disagree: 12
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4. Inferential Statistical Analysis

To validate the observed patterns, rigorous statistical tests were conducted to determine the significance of these relationships.

A. Preference vs. Stress (T-Test)

An Independent Samples T-test was performed to see if the **Preferred System Type** (MCQ vs. Written) significantly impacts overall stress levels.

- **Results:** T-statistic: **1.094**, P-value: **0.278**.
- **Conclusion:** The difference is **not statistically significant** ($P > 0.05$). This suggests that while students have a preference, the inherent stress of the academic process affects both groups similarly.

B. Gender & Field Independence (Chi-Square Test)

We tested if **Gender** or **Academic Field** dictates a student's preferred exam system.

- **Gender Results:** Chi-Square: **7.046**, P-value: **0.133**.
- **Field Results:** Chi-Square: **28.83**, P-value: **0.421**.
- **Conclusion:** In both cases, the results were **not significant**. This proves that exam preference is an individual psychological choice and is not biased by gender or the student's area of study.

5. Predictive Regression Analysis

We built an **OLS Regression Model** to understand the specific drivers of **Written Exam Time Stress**.

Model Performance:

- **R-squared: 0.298** (The model explains ~30% of the variance in time-related stress).
- **Prob (F-statistic): 2.46e-13** (Highly significant model).

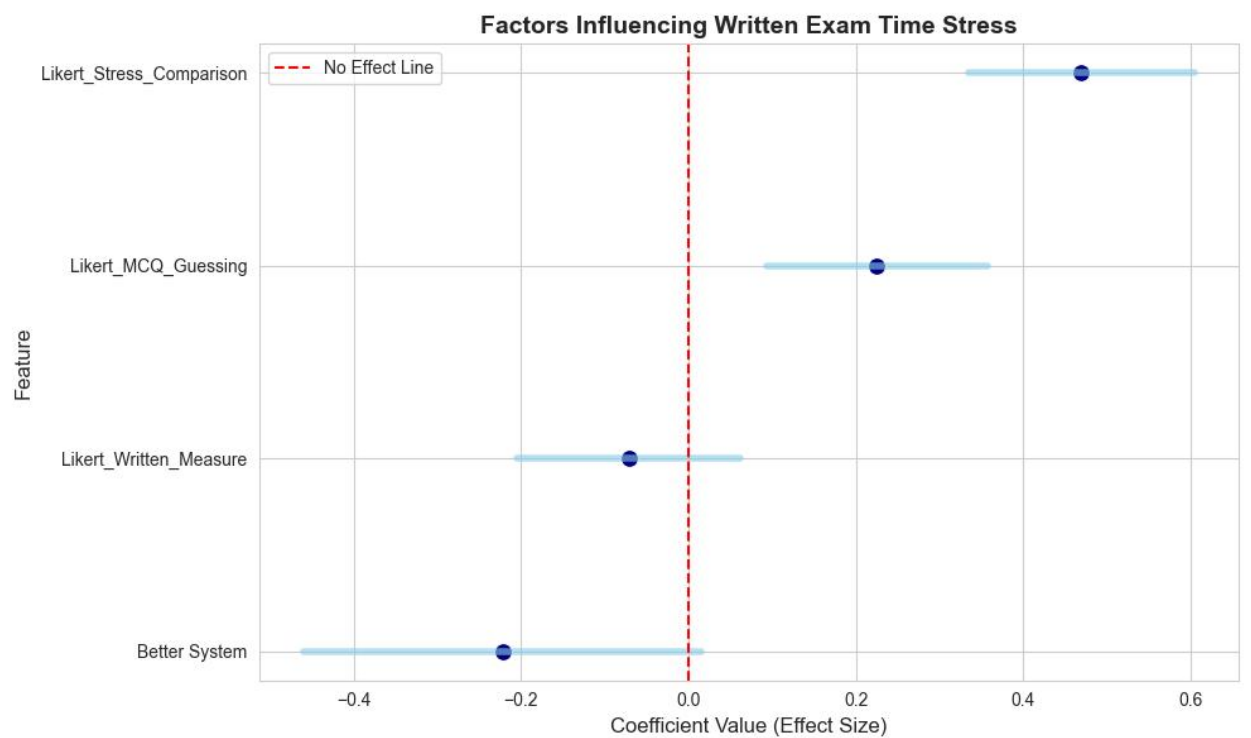
Key Drivers of Stress:

Factor	Coefficient	P-Value	Significance
Stress Comparison	0.4695	0.000	Critical Impact
MCQ Guessing	0.2254	0.001	High Impact
Better System Belief	-0.2224	0.066	Marginal Impact

Insight: The analysis reveals that the **reliance on "Guessing"** in MCQs is a major predictor of stress in written exams. Students who use guessing as a strategy find the transition to written formats—where depth is required—highly stressful.

6. Visualization of Influence Factors

The following chart visualizes the coefficients of our regression model, highlighting the factors that most significantly drive student stress.



Final Conclusion

The research findings successfully met the project objectives. It is concluded that while students value the depth of written exams, the high stress levels associated with them—primarily due to time pressure—drive a majority preference toward MCQ formats.

Statistical validation further clarifies that this stress is not tied to gender or field, but is deeply connected to student strategies, specifically the habit of **guessing in MCQs**. Addressing academic well-being requires a balanced approach that mitigates time pressure and reinforces the qualitative benefits of written assessments.